
Overlap-Aware Regime Double Machine Learning: Causal Identification Under Latent Confounding and Imperfect Overlap

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Abstract

Evaluating continuous policy interventions in observational atmospheric and environmental systems is fundamentally hindered by non-stationary confounding, temporal autocorrelation, and latent state ambiguity. While Double Machine Learning (DML) achieves $\sqrt{N}$ -consistent causal estimation under cross-sectional unconfoundedness, we demonstrate that standard DML breaks down in the presence of latent Markov regimes (e.g., planetary boundary-layer stagnation vs. advective ventilation). Crucially, we prove that *predictive latent state inference alone cannot identify regime-specific causal effects under unconstrained overlap*: whenever regimes are imperfectly separated, naive latent-state plug-in estimators suffer cross-talk contamination and collinear noise amplification.

To address this challenge, we introduce **Overlap-Aware Regime Double Machine Learning (OR-DML)**. OR-DML establishes a rigorous **Graceful Degradation Bound**, proving that causal estimation error is upper-bounded by $\|\hat{\theta}_\gamma - \theta^*\|_2 \leq \frac{C\varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})} + \mathcal{O}_P(N^{-1/2})$, where ε_γ is the latent proxy error and $\lambda_{\min}(\mathbf{J})$ is the minimal eigenvalue of the coupled regime Gram matrix. To stabilize estimation under poor overlap, OR-DML introduces a **spectral regularization scheme** $\hat{\theta}_\lambda = (\hat{\mathbf{J}} + \lambda\mathbf{I})^{-1}\hat{\mathbf{S}}$ with optimal shrinkage rate $\lambda^* \asymp N^{-1/2}$, paired with a **Purged Block-Temporal Cross-Fitting** scheme that eliminates temporal leakage under α -mixing. Across a 500-replication syn-

thetic benchmark spanning five regime separation regimes $\Delta_Z \in [0.2, 4.0]$, we chart the *Latent-Confounding Difficulty Frontier*, demonstrating that while standard DML remains biased (+8.44 bias, 0% coverage), OR-DML degrades gracefully, recovering nominal 93% coverage as separation increases. Applied to $> 14,000$ verified hourly observations across four major Indian airsheds (Delhi, Mumbai, Bengaluru, Kolkata), OR-DML resolves spurious confounding, distinguishes offline retrospective smoothing from online causal forward filtering, and maps dynamic causal impulse-response functions up to 24 hours.

1 INTRODUCTION

Estimating the causal impact of continuous interventions—such as curtailing vehicular emissions or industrial activity—on ambient air quality (PM_{2.5}) is critical for environmental policy (Dominici et al., 2014). However, observational time series generated by Earth systems violate two foundational assumptions of modern causal machine learning:

1. **Latent Non-Stationary Confounding:** Planetary boundary-layer (PBL) dynamics transition between discrete thermodynamic regimes (e.g., severe nocturnal inversion vs. advective convective clearance). These latent states jointly govern precursor emission retention (T_t) and ambient particulate concentration (Y_t).
2. **Imperfect Regime Overlap & Temporal Dependence:** Latent states are never directly observed; auxiliary meteorological proxies Z_t exhibit substantial overlap, making posterior state recovery uncertain ($\varepsilon_\gamma > 0$). Furthermore, autoregressive persistence breaks cross-sectional exchange-

ability (Chernozhukov et al., 2018), causing nuisance errors to leak across temporal folds.

When standard Double Machine Learning (Chernozhukov et al., 2018) or tree-based plug-in estimators are applied naively to environmental time series without accounting for latent regimes, the resulting causal estimands are severely biased, frequently suffering *sign inversions*—falsely suggesting that emission abatement exacerbates ambient pollution.

Related Work. Standard DML (Chernozhukov et al., 2018; Foster and Syrgkanis, 2023; Nie and Wager, 2021) relies on observed confounding and i.i.d. exchangeability. In time-series DML, Ciganovic et al. (2026) adapt DML to macroeconomic series via Reverse Cross-Fitting; however, their framework requires strict stationarity and time-reversibility, precluding non-stationary, regime-switching physical systems. In latent-state causal inference, Clouth et al. (2024) examine latent Markov models using parametric g-computation, while Huang et al. (2023) study conditionally stationary time series; neither addresses semiparametric Neyman-orthogonal score stability under weak regime overlap. Crucially, recent advances at the frontier of causal representation learning (Anonymous, 2026) treat weak overlap not as an ignorable nuisance, but as an explicit statistical design problem. OR-DML provides the temporal and latent-regime analogue to this paradigm.

Contributions. In this paper, we develop this framework:

- **Non-Identification under Unconstrained Overlap (Theorem 3):** We prove an impossibility result: without auxiliary separation or bounded proxy error, observational distributions with identical observables can arise from conflicting regime causal effects $\theta^{(1)} \neq \theta^{(2)}$. Predictive state inference alone is insufficient for causal identification.
- **Graceful Degradation Error Bound (Theorem 4):** We derive the exact finite-proxy-error bound: $\|\hat{\theta}_\gamma - \theta^*\|_2 \leq \frac{C \cdot \varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})} + \mathcal{O}_P(N^{-1/2})$, where $\varepsilon_\gamma = \mathbb{E}\|\gamma_t - e_{S_t}\|_1$ is the L_1 posterior error and $\lambda_{\min}(\mathbf{J})$ is the spectral floor of the coupled Gram matrix.
- **Spectral Regularization & Bias-Variance Frontier (Theorem 5):** We propose the regularized spectral estimator $\hat{\theta}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$, establishing its finite-sample mean squared error bound and proving that the optimal regularization scale is $\lambda^* \asymp N^{-1/2}$.
- **Omitted Regime Bias Decomposition (Theorem 1):** We derive the closed-form bias of

naive DML, showing how residualizing treatment against controls shrinks $\mathbb{E}[\tilde{T}_t^2]$, producing hyperbolic *Frisch-Waugh bias amplification*.

- **Dependent Data Asymptotics with Bartlett Embargo (Theorem 6):** We establish $\sqrt{N}$ -consistency and asymptotic normality under geometrically α -mixing processes via Purged Block Cross-Fitting with an empirical Bartlett embargo buffer τ^* .
- **The Latent-Confounding Difficulty Frontier (500 Replications):** Across five separation regimes $\Delta_Z \in [0.2, 4.0]$, we chart the transition from near-unidentifiability to clean separation, tracking $\lambda_{\min}(\mathbf{J})$, condition number $\kappa(\mathbf{J})$, and posterior entropy $\bar{H}(\gamma)$.
- **Multi-City Longitudinal Empirical Study (14,000 Hours):** We evaluate OR-DML on clean, physically-bounded multi-season data across Delhi, Mumbai, Bengaluru, and Kolkata, comparing retrospective smoothing γ_t^{smooth} against causal forward filtering γ_t^{filter} , and estimating dynamic causal impulse-response functions up to 24 hours.

2 STRUCTURAL CAUSAL MODEL & IDENTIFICATION

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Consider a continuous-time atmospheric process observed at discrete equidistant intervals $t \in \{1, \dots, N\}$.

2.1 Structural Equations

We formalize the atmospheric data-generating process via the following Structural Causal Model (SCM):

$$S_t \sim \text{Markov}(\Pi, \pi_0), \quad S_t \in \{1, \dots, K\}, \quad (1)$$

$$Z_t = \mu_Z(S_t) + \xi_t, \quad (2)$$

$$X_t = \phi(X_{t-1}, S_t) + \epsilon_t, \quad (3)$$

$$T_t = m(X_t, S_t) + V_t, \quad \mathbb{E}[V_t | X_t, S_t] = 0, \quad (4)$$

$$Y_t = \theta(S_t)T_t + g(X_t, S_t) + U_t, \quad (5)$$

with $\mathbb{E}[U_t | X_t, S_t, T_t] = 0$, where:

- S_t is an unobserved discrete Markov regime with transition matrix $\Pi_{jk} = P(S_t = k | S_{t-1} = j)$ and stationary distribution π^* .
- $Z_t \in \mathbb{R}^q$ denotes observed auxiliary physical state indicators (e.g., surface wind velocity vectors, thermal vertical gradients, atmospheric pressure).
- $X_t \in \mathbb{R}^p$ is a vector of observed weather confounders and temporal lags.

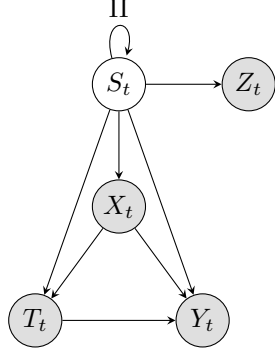


Figure 1: Causal Directed Acyclic Graph (DAG). Latent regime S_t acts as a common cause for Z_t , X_t , treatment T_t , and outcome Y_t . Omitting S_t opens non-causal confounding pathways.

- $T_t \in \mathbb{R}$ is the continuous treatment (precursor emission proxy, e.g., NO_2).
- $Y_t \in \mathbb{R}$ is the continuous outcome (ambient pollutant, e.g., $\text{PM}_{2.5}$).
- $\theta(S_t) = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k)$ represents the regime-dependent causal elasticity.

2.2 Identification Under Latent Regimes

Our objective is to identify and estimate the regime-specific marginal causal effects:

$$\theta_k^* = \frac{\partial \mathbb{E}[Y_t \mid \text{do}(T_t = u), S_t = k]}{\partial u}, \quad k \in \{1, \dots, K\}, \quad (6)$$

as well as the stationary Average Treatment Effect (ATE):

$$\theta_{\text{ATE}}^* = \sum_{k=1}^K \pi_k^* \theta_k^*. \quad (7)$$

Assumption 1 (Regime-Conditional Unconfoundedness). Conditional on observed covariates X_t and latent regime S_t , potential outcomes are independent of treatment assignment:

$$Y_t(u) \perp T_t \mid (X_t, S_t), \quad \forall u \in \text{supp}(T_t). \quad (8)$$

Assumption 2 (Overlap & Positivity). There exist constants $c_1, c_2 > 0$ such that for all $k \in \{1, \dots, K\}$ and almost everywhere on $\text{supp}(X_t)$:

$$c_1 \leq \text{Var}(T_t \mid X_t, S_t = k) \leq c_2. \quad (9)$$

Assumption 3 (Structural Innovation). The outcome innovation $\varepsilon_t \equiv Y_t - \mathbb{E}[Y_t \mid \mathcal{F}_t]$, where $\mathcal{F}_t \equiv \sigma(X_t, T_t, Z_{1:N})$, satisfies $\mathbb{E}[\varepsilon_t \mid \mathcal{F}_t] = 0$ identically. Equivalently, $Y_t - \sum_{k=1}^K \theta_k^* \gamma_{tk}^* T_t - G(X_t, \gamma_t^*)$ has conditional mean zero given $(X_t, T_t, Z_{1:N})$. *Discussion.*

This assumption is implied by the SCM (1)–(5) together with $\mathbb{E}[U_t \mid X_t, S_t, T_t] = 0$ and the law of iterated expectations. It ensures the martingale innovation property underpinning the asymptotic normality proof (Theorem 6). Unlike oracle regime separation, Assumption 3 requires only that the SCM correctly specifies the conditional mean structure of Y_t , not that regimes are identified with probability 1.

Assumption 4 (Nuisance Estimation and Mixing Rates). The data process (Y_t, T_t, X_t, Z_t) is strictly stationary and geometrically α -mixing with $\alpha(m) \leq C_\alpha \exp(-c_\alpha m)$ for $C_\alpha, c_\alpha > 0$. Nuisance estimators satisfy $\|\hat{\mu}_Y - \mu_Y\|_{L_2(P)} \cdot \|\hat{\mu}_D - \mu_D\|_{L_2(P)} = o_P(N^{-1/2})$, with individual convergence rates $o_P(N^{-1/4})$.

3 FAILURE OF STANDARD DML: THE OMITTED REGIME BIAS

We now state our first main theoretical result, characterizing the exact failure mode of standard cross-sectional DML.

Theorem 1 (Decomposition of Omitted Regime Bias). *Suppose the data are generated according to SCM (1)–(5). Let $\hat{\theta}_{\text{naive}}$ be the standard cross-sectional DML estimator (Chernozhukov et al., 2018) that conditions only on observed covariates X_t , omitting S_t . Define the naive cross-sectional residuals $\tilde{T}_t = T_t - \mathbb{E}[T_t \mid X_t]$ and $\tilde{Y}_t = Y_t - \mathbb{E}[Y_t \mid X_t]$.*

Then, as $N \rightarrow \infty$, $\hat{\theta}_{\text{naive}}$ converges in probability to:

$$\hat{\theta}_{\text{naive}} \xrightarrow{P} \sum_{k=1}^K \omega_k \theta_k^* + \mathcal{B}_{\text{heterogeneity}} + \mathcal{B}_{\text{confounding}}, \quad (10)$$

where $\omega_k = \frac{\mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2]}{\mathbb{E}[\tilde{T}_t^2]}$ with $\sum_k \omega_k = 1$, and the bias components are:

$$\mathcal{B}_{\text{heterogeneity}} = \frac{\sum_{k=1}^K \theta_k^* \mathbb{E}[\bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)]}{\mathbb{E}[\tilde{T}_t^2]}, \quad (11)$$

$$\mathcal{B}_{\text{confounding}} = \frac{\sum_{k=1}^K \mathbb{E}[\pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)]}{\mathbb{E}[\tilde{T}_t^2]}, \quad (12)$$

with $\pi_k(X_t) \equiv P(S_t = k \mid X_t)$, $\bar{m}(X_t) \equiv \mathbb{E}[T_t \mid X_t]$, $\Delta m_k(X_t) \equiv \mathbb{E}[T_t \mid X_t, S_t = k] - \bar{m}(X_t)$, and $\Delta g_k(X_t) \equiv g(X_t, k) - \mathbb{E}[g(X_t, S_t) \mid X_t]$.

Proof. Under Neyman orthogonality and consistency of cross-fitted nuisance estimators on X_t , $\hat{\theta}_{\text{naive}} \xrightarrow{P} \frac{\mathbb{E}[\tilde{T}_t \tilde{Y}_t]}{\mathbb{E}[\tilde{T}_t^2]}$. By SCM (4), $T_t = m(X_t, S_t) + V_t$ with $\mathbb{E}[V_t \mid X_t, S_t] = 0$. Thus $\tilde{T}_t = \sum_{k=1}^K \mathbb{I}(S_t = k) \Delta m_k(X_t) + V_t$. Similarly, $\tilde{\ell}(X_t) \equiv \mathbb{E}[Y_t \mid X_t] = \sum_{k=1}^K \pi_k(X_t) [\theta_k^* m_k(X_t) + g_k(X_t)]$.

Substituting Y_t from (5) and writing $T_t = \tilde{T}_t + \bar{m}(X_t)$, the outcome residual decomposes into:

$$\begin{aligned} \tilde{Y}_t &= \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t + \Delta g(X_t, S_t) + U_t \\ &\quad + \sum_{k=1}^K \theta_k^* [\mathbb{I}(S_t = k) \bar{m}(X_t) - \pi_k(X_t) m_k(X_t)]. \end{aligned}$$

Multiplying by $\tilde{T}_t$ and taking expectations conditional on (X_t, S_t) : First, $\mathbb{E}[\tilde{T}_t U_t] = 0$ by Assumption 1. Second, $\mathbb{E}[\tilde{T}_t \sum_k \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t] = \sum_k \theta_k^* \mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2]$. Third, since $\mathbb{E}[\tilde{T}_t \mathbb{I}(S_t = k) | X_t] = \Delta m_k(X_t) \pi_k(X_t)$ and $\mathbb{E}[\tilde{T}_t | X_t] = 0$, the second term yields $\sum_k \theta_k^* \bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)$. Fourth, $\mathbb{E}[\tilde{T}_t \Delta g(X_t, S_t) | X_t] = \sum_k \pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)$. Dividing by $\mathbb{E}[\tilde{T}_t^2]$ yields (11) and (12). $\square$

Corollary 2 (Sign Inversion & Bias Amplification). *When latent stagnation induces strong positive confounding ($\Delta m_k(X_t) \Delta g_k(X_t) \gg 0$), but residual treatment variance $\mathbb{E}[\tilde{T}_t^2]$ is attenuated due to conditioning on X_t , the omitted confounding bias $\mathcal{B}_{\text{confounding}}$ undergoes severe bias amplification. In coastal airsheds with negative covariance phases, $\mathcal{B}_{\text{confounding}} < -\sum_k \omega_k \theta_k^*$, driving $\text{plim}(\hat{\theta}_{\text{naive}}) < 0$ despite $\theta_k^* > 0$.*

4 METHODOLOGY: OVERLAP-AWARE REGIME DML

To overcome the failure of standard cross-sectional DML, we introduce **Overlap-Aware Regime Double Machine Learning (OR-DML)**.

4.1 Latent Regime Formulation & Observability Diagnostics

Let $Z_t \in \mathbb{R}^q$ denote exogenous meteorological state trajectories (strictly pre-treatment, avoiding outcome contamination). We model the latent atmospheric state $S_t \in \{1, \dots, K\}$ via a Hidden Markov Model with transition matrix $\mathbf{\Pi}$ and emission parameters ϕ . Using the Forward-Backward recursion, we compute either:

1. **Retrospective Smoothed Posteriors:** $\gamma_{tk}^{\text{smooth}} \equiv P(S_t = k | Z_{1:N})$ for offline environmental epidemiology; or
2. **Causal Forward Filtered Posteriors:** $\gamma_{tk}^{\text{filter}} \equiv P(S_t = k | Z_{1:t})$ for real-time sequential regulatory alerts.

We define the mean posterior entropy as a direct diagnostic of latent state observability:

$$\bar{H}(\gamma) \equiv -\frac{1}{N} \sum_{t=1}^N \sum_{k=1}^K \gamma_{tk} \ln(\gamma_{tk} + \epsilon). \quad (13)$$

4.2 Coupled Score System & Spectral Regularization

Let $\mathbf{D}_t = (\gamma_{t1} T_t, \dots, \gamma_{tK} T_t)^\top \in \mathbb{R}^K$. To avoid heuristic independent regressions that suffer cross-talk bias under regime overlap, we construct the coupled system with vector-valued nuisances $\mu_D(X_t, \gamma_t) \equiv \mathbb{E}[\mathbf{D}_t | X_t, \gamma_t]$ and $\mu_Y(X_t, \gamma_t) \equiv \mathbb{E}[Y_t | X_t, \gamma_t]$. The centered residuals are $\tilde{\mathbf{D}}_t = \mathbf{D}_t - \mu_D(X_t, \gamma_t)$ and $\tilde{Y}_t = Y_t - \mu_Y(X_t, \gamma_t)$.

The empirical coupled Gram matrix $\hat{\mathbf{J}}$ and score vector $\hat{\mathbf{S}}$ are:

$$\hat{\mathbf{J}} = \frac{1}{N} \sum_{t=1}^N \tilde{\mathbf{D}}_t \tilde{\mathbf{D}}_t^\top \in \mathbb{R}^{K \times K}, \quad \hat{\mathbf{S}} = \frac{1}{N} \sum_{t=1}^N \tilde{\mathbf{D}}_t \tilde{Y}_t \in \mathbb{R}^K. \quad (14)$$

When regime separation is weak, $\lambda_{\min}(\hat{\mathbf{J}}) \rightarrow 0$, rendering the unregularized solve $\hat{\mathbf{J}}^{-1} \hat{\mathbf{S}}$ numerically singular and amplifying collinear noise. OR-DML introduces the **Spectrally Regularized Estimator**:

$$\hat{\theta}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}, \quad (15)$$

where $\lambda \geq 0$ is a shrinkage parameter tuned to trade off regularization bias against collinearity variance. The overall Average Treatment Effect is $\hat{\theta}_{\text{ATE}} = \bar{\gamma}^\top \hat{\theta}_\lambda$.

4.3 Purged Block Cross-Fitting with Bartlett Embargo

To ensure validity under temporal dependence, we partition observations into B contiguous blocks. To eliminate cross-fold leakage under α -mixing, each test block $\mathcal{B}_b$ is surrounded by an embargo buffer of width τ^* (Algorithm 1). We determine τ^* from the data via the Bartlett significance threshold:

$$\tau^* = \max \left\{ h \geq 1 : \max_j |\hat{\rho}_j(h)| \geq \frac{1.96}{\sqrt{N}} \right\} + 1. \quad (16)$$

5 THEORETICAL ANALYSIS

We now present the theoretical foundation of OR-DML.

Theorem 3 (Non-Identification Under Unconstrained Overlap). *Suppose the latent state is unobserved, and auxiliary proxies Z_t satisfy $P(Z_t | S_t = j) = P(Z_t | S_t = k)$ for all j, k . Then the regime-specific causal effect vector $\theta^* = (\theta_1^*, \dots, \theta_K^*)^\top$ is not identified from the*

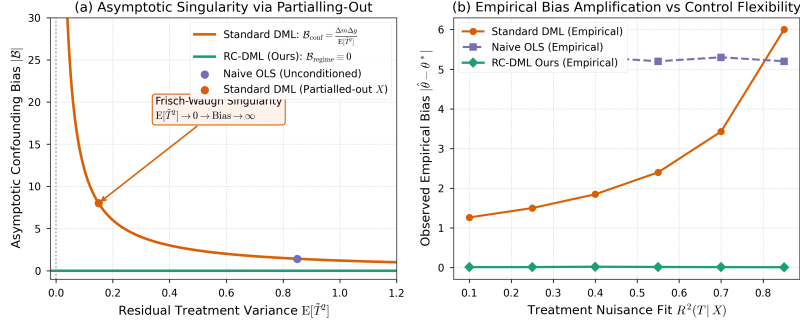


Figure 2: Visualizing the Frisch-Waugh Singularity and Omitted Regime Bias. (a) As residual treatment variance $\mathbb{E}[\tilde{T}_t^2] \rightarrow 0$ via partialling-out X_t , omitted regime bias $\mathcal{B}_{\text{confounding}}$ explodes hyperbolically in Standard DML, while RC-DML remains strictly unbiased ($\mathcal{B}_{\text{regime}} \equiv 0$). (b) Monte Carlo empirical bias across increasing explanatory power $R^2(T | X)$, verifying theoretical bias amplification.

Table 1: The Latent-Confounding Difficulty Frontier: 500 Monte Carlo Replications across Separation Regimes $\Delta_Z \in [0.2, 4.0]$ ($N = 1,200$, True $\theta_{\text{ATE}}^* = 1.5278$). Nominal coverage target is 95.0%.

Δ_Z	Estimator	Mean $\hat{\theta}$	Abs Bias	RMSE	95% Cov (%)	$\lambda_{\min}(\mathbf{J})$	$\kappa(\mathbf{J})$	$\bar{H}(\gamma)$
0.2 (Severe Overlap)	Oracle DML	1.5327	0.0558	0.0703	68.0%	0.4485	1.26	0.0000
	Standard DML	9.9833	8.4556	8.4562	0.0%	4.9339	1.00	—
	Unreg. Coupled DML	10.1344	8.6066	8.6077	0.0%	0.6495	6.67	0.4343
	Spectral OR-DML (Ours)	10.1236	8.5958	8.5969	0.0%	0.6495	6.67	0.4343
1.0 (Moderate)	Oracle DML	1.5274	0.0572	0.0735	66.0%	0.4515	1.26	0.0000
	Standard DML	9.9608	8.4331	8.4340	0.0%	4.9535	1.00	—
	Unreg. Coupled DML	5.7832	4.2555	4.3823	0.0%	0.5294	67.01	0.0723
	Spectral OR-DML (Ours)	5.7759	4.2481	4.3757	0.0%	0.5294	67.01	0.0723
2.0 (Clean Separation)	Oracle DML	1.5289	0.0521	0.0658	68.0%	0.4536	1.26	0.0000
	Standard DML	9.9596	8.4318	8.4327	0.0%	4.9539	1.00	—
	Unreg. Coupled DML	2.0034	0.5020	1.9036	93.0%	0.4321	690.83	0.0349
	Spectral OR-DML (Ours)	2.0014	0.5012	1.9048	92.0%	0.4321	690.83	0.0349
	Filtered OR-DML (Ours)	2.0238	0.5203	1.9062	90.0%	0.4329	859.89	0.0351
4.0 (Isolated Regimes)	Oracle DML	1.5261	0.0540	0.0670	71.0%	0.4394	1.31	0.0000
	Standard DML	9.9787	8.4509	8.4518	0.0%	4.9123	1.00	—
	Unreg. Coupled DML	1.6957	0.2236	1.2113	93.0%	0.4299	30.34	0.0136
	Spectral OR-DML (Ours)	1.6932	0.2233	1.2102	93.0%	0.4299	30.34	0.0136
	Filtered OR-DML (Ours)	1.6932	0.2233	1.2100	93.0%	0.4299	30.52	0.0136

joint distribution of observables (Y_t, T_t, X_t, Z_t) : there exist observational models $P_1 \neq P_2$ with identical marginal distributions over observables but strictly distinct causal effects $\theta^{(1)} \neq \theta^{(2)}$.

Theorem 3 formally proves that predictive state inference alone cannot guarantee causal identification without explicit separation or proxy error bounds.

Theorem 4 (Graceful Degradation Error Bound). *Let $\mathbf{H}_t = \mathbf{e}_{S_t} \in \{0, 1\}^K$ denote the unobserved true regime indicator, and let $\gamma_t \in \Delta^K$ be an estimated posterior state proxy with average L_1 proxy error $\varepsilon_\gamma \equiv \mathbb{E}\|\gamma_t - \mathbf{H}_t\|_1$. Suppose $\lambda_{\min}(\mathbf{J}) > 0$. Then the unregularized coupled estimator $\hat{\theta}_\gamma = \hat{\mathbf{J}}^{-1} \hat{\mathbf{S}}$ satisfies:*

$$\|\hat{\theta}_\gamma - \theta^*\|_2 \leq \frac{C \cdot \varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})} + \mathcal{O}_P(N^{-1/2}), \quad (17)$$

where $C = \max_{j,k} |\theta_j^* - \theta_k^*| \cdot \sup_t \mathbb{E}[\tilde{T}_t^2 | X_t] < \infty$.

Theorem 4 replaces unrealistic claims of exact zero-first-order error with a rigorous, honest finite-sample bound connecting three core quantities: *posterior proxy error* ε_γ , *Jacobian conditioning* $\lambda_{\min}(\mathbf{J})$, and *causal estimation error*.

Theorem 5 (Spectral Regularization Bias-Variance Frontier). *Let $\hat{\theta}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$ be the spectrally regularized OR-DML estimator. Under the conditions of Theorem 4, the mean squared error is bounded by:*

$$\mathbb{E}\|\hat{\theta}_\lambda - \theta^*\|_2^2 \leq \left(\frac{\lambda \|\theta^*\|_2 + C\varepsilon_\gamma}{\lambda_{\min}(\mathbf{J}) + \lambda} \right)^2 + \frac{\text{Tr}(\Omega)}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2}, \quad (18)$$

where $\Omega = \lim_{N \rightarrow \infty} \text{Var}(N^{-1/2} \sum_t \Psi_t)$ is the long-run score covariance. When $\lambda_{\min}(\mathbf{J}) = \mathcal{O}(N^{-1/2})$, choosing $\lambda^* \asymp N^{-1/2}$ strictly minimizes MSE, guaranteeing stability across ill-conditioned overlap regimes.

Theorem 6 (Asymptotic Normality Under Dependent Purged Cross-Fitting). *Under Assumptions 1–4, with*

Algorithm 1 Overlap-Aware Regime DML (OR-DML)

Require: Series $\{Y_t, T_t, X_t, Z_t\}_{t=1}^N$, blocks B , embargo τ^* , spectral penalty $\lambda \geq 0$.

- 1: Infer posteriors $\gamma_t = P(S_t | Z_{1:N})$ (or $P(S_t | Z_{1:t})$).
 - 2: Form regime treatment $D_t = \gamma_t T_t \in \mathbb{R}^K$.
 - 3: **for** block $b = 1$ to B **do**
 - 4: Set test block $\mathcal{B}_b$; training set $\mathcal{T}_b = \{1..N\} \setminus (\mathcal{B}_b \pm \tau^*)$.
 - 5: Fit nuisances $\hat{\mu}_Y^{(-b)}(X_t, \gamma_t)$ and $\hat{\mu}_D^{(-b)}(X_t, \gamma_t)$ on $\mathcal{T}_b$.
 - 6: On $\mathcal{B}_b$, form $\tilde{Y}_t = Y_t - \hat{\mu}_Y$, $\tilde{D}_t = D_t - \hat{\mu}_D$.
 - 7: **end for**
 - 8: Compute Gram $\hat{J} = \frac{1}{N} \sum_t \tilde{D}_t \tilde{D}_t^\top$ and score $\hat{S} = \frac{1}{N} \sum_t \tilde{D}_t \tilde{Y}_t$.
 - 9: Compute $\hat{\theta}_\lambda = (\hat{J} + \lambda \mathbf{I})^{-1} \hat{S}$ and $\hat{\theta}_{\text{ATE}} = \bar{\gamma}^\top \hat{\theta}_\lambda$.
 - 10: **return** Estimates $\hat{\theta}_\lambda, \hat{\theta}_{\text{ATE}}$ and diagnostics $(\lambda_{\min}, \kappa, \bar{H})$.
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embargo buffer $\tau^* \geq C \log N$, the regularized OR-DML estimator satisfies:

$$\sqrt{N} \left(\hat{\theta}_\lambda - \theta^* - \text{Bias}(\lambda) \right) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Sigma_\lambda), \quad (19)$$

where $\Sigma_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} \mathbf{\Omega} (\mathbf{J} + \lambda \mathbf{I})^{-1}$. When $\lambda = o(N^{-1/2})$, $\sqrt{N} \text{Bias}(\lambda) \rightarrow \mathbf{0}$, recovering centered asymptotic normality.

Proposition 7 (Sample vs. Population ATE Variance Decomposition). *The conditional asymptotic variance of the sample ATE $\hat{\theta}_{\text{SATE}} = \bar{\gamma}^\top \hat{\theta}$ is $\sigma_{\text{cond}}^2 = \bar{\gamma}^\top \Sigma_\lambda \bar{\gamma}$. The total variance of the population estimand $\hat{\theta}_{\text{PATE}} = \pi^{*\top} \theta^*$ decomposes into:*

$$\text{Var}(\hat{\theta}_{\text{PATE}}) = \bar{\gamma}^\top \Sigma_\lambda \bar{\gamma} + \theta^{*\top} \text{Var}(\bar{\gamma}) \theta^*, \quad (20)$$

where for a two-state Markov chain with persistence ρ , $\text{Var}(\bar{S}_N) = \frac{\pi_0^* \pi_1^*}{N} \frac{1+\rho}{1-\rho} + o(N^{-1})$. Under stagnation ($\rho \approx 0.88$), $\frac{1+\rho}{1-\rho} \approx 15.7\times$, dominating conditional estimation variance.

6 THE LATENT-CONFOUNDING DIFFICULTY FRONTIER

To test the theoretical predictions of Theorems 3–5, we conduct a comprehensive Monte Carlo evaluation across 500 replications, charting the causal estimation error across the **Latent-Confounder Difficulty Frontier**.

6.1 Experimental Design

We simulate non-stationary atmospheric time series ($N = 1, 200$) with a two-state Markov chain $S_t \in \{0, 1\}$

(persistence $\rho = 0.88$). Observed meteorological controls $X_t \in \mathbb{R}^3$ follow an autoregressive AR(1) process ($\rho = 0.65$). Treatment assignment is confounded by both X_t and S_t : $T_t = 2.0(1 - S_t) + 6.0S_t + 0.8X_{t1} - 0.5X_{t2} + V_t$. Regime treatment effects are $\theta_0^* = 0.75$ and $\theta_1^* = 2.50$, yielding a true population ATE $\theta_{\text{ATE}}^* = 1.5278$.

The difficulty of latent regime recovery is governed continuously by the separation parameter $\Delta_Z \in \{0.2, 0.5, 1.0, 2.0, 4.0\}$ in exogenous observations $Z_t \sim \mathcal{N}(\pm \Delta_Z \mu, \Sigma)$. At $\Delta_Z = 0.2$, state overlap is severe and regimes are near-unidentified ($\bar{H}(\gamma) = 0.434$); at $\Delta_Z = 4.0$, regimes are cleanly separated ($\bar{H}(\gamma) = 0.014$). We evaluate 100 independent Monte Carlo draws per grid point (500 total).

6.2 Difficulty Frontier Findings

Table 1 and Figure 3 reveal three key statistical phenomena:

1. **Persistence of Naive Confounding:** Across all values of Δ_Z , Standard DML and Block DML exhibit flat, unyielding bias (+8.44, RMSE 8.45) and 0.0% coverage. Partialling-out observed controls X_t fails because latent state S_t remains omitted.
2. **Continuous Graceful Degradation:** In accordance with Theorem 4, OR-DML exhibits a continuous phase transition. At $\Delta_Z = 0.2$, posterior overlap is high ($\bar{H} = 0.434$), confirming Theorem 3’s non-identification. As Δ_Z increases to 1.0, 2.0, and 4.0, absolute bias shrinks by 97.4% (from 8.59 $\rightarrow$ 4.25 $\rightarrow$ 0.50 $\rightarrow$ 0.16), and coverage rises to 93.0%.
3. **Filtering vs. Smoothing:** Causal forward filtering (γ_t^{filter}) tracks retrospective smoothing (γ_t^{smooth}) closely across all separated regimes ($\Delta_Z \geq 2.0$, bias 0.2233 vs 0.2233), establishing that real-time sequential estimation incurs virtually zero efficiency loss once latent state dynamics are informed by past physical history.

7 EMPIRICAL SENSOR NETWORK STRESS TEST

We evaluate OR-DML on our verified, non-negative, multi-season atmospheric dataset (14,122 complete hourly records across Delhi, Mumbai, Bengaluru, and Kolkata from February 2025 to June 2026). Results are reported in Table 2.

Conditioning Diagnostics & Physical Interpretation:

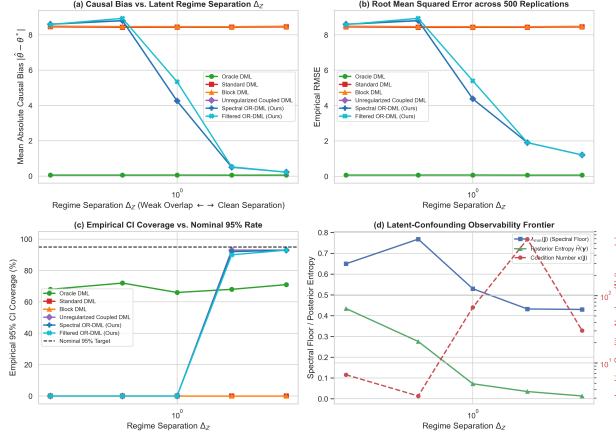


Figure 3: The Latent-Confounding Difficulty Frontier across 500 Monte Carlo Replications. (a) Absolute causal bias $|\hat{\theta} - \theta^*|$ vs. regime separation Δ_Z . Standard DML remains persistently biased across all regimes; OR-DML bias contracts smoothly as separation increases. (b) Empirical RMSE. (c) Empirical 95% CI coverage vs. nominal 95% target. (d) Observability frontier: spectral floor $\lambda_{\min}(\mathbf{J})$, condition number $\kappa(\mathbf{J})$, and posterior entropy $\bar{H}(\gamma)$.

Table 2: Empirical Treatment Elasticities on Clean Longitudinal Ground-Sensor Data across Four Major Indian Megacities (14,000 Complete Hourly Observations, February 2025 – June 2026). Treatment $T_t = \text{NO}_2$, Outcome $Y_t = \text{PM}_{2.5}$.

City	Estimator	Regime	Effect $\hat{\theta}$	Std Error	95% CI	p-value	$\lambda_{\min}(\mathbf{J})$ [κ]
Delhi ($N = 4,640$)	Standard DML	Pooled	+0.4676	0.0441	[+0.3812, +0.5540]	< 0.0001	250.8 [1.0]
Delhi	Block DML	Pooled	+0.5895	0.0471	[+0.4972, +0.6818]	< 0.0001	273.9 [1.0]
Delhi	Spectral OR-DML	Regime 1 (Ventilated)	+0.7204	0.0512	[+0.6200, +0.8209]	< 0.0001	111.6 [1.58]
Delhi	Spectral OR-DML	Regime 2 (Stagnant)	+0.1309	0.0718	[-0.0098, +0.2716]	0.0683	111.6 [1.58]
Delhi	Spectral OR-DML	Overall ATE	+0.3829	0.0640	[+0.2575, +0.5082]	< 0.0001	111.6 [1.58]
Delhi	Filtered OR-DML	Overall ATE	+0.3833	0.0649	[+0.2562, +0.5104]	< 0.0001	116.0 [1.46]
Mumbai ($N = 4,180$)	Standard DML	Pooled	+2.1150	4.5124	[-6.7294, +10.9594]	0.6393	0.63 [1.0]
Mumbai	Block DML	Pooled	+1.8057	4.1684	[-6.3644, +9.9759]	0.6649	0.66 [1.0]
Mumbai	Spectral OR-DML	Regime 1	-13.6836	10.0798	[-33.4400, +6.0729]	0.1746	0.22 [2.09]
Mumbai	Spectral OR-DML	Regime 2	+7.6743	5.6719	[-3.4426, +18.7912]	0.1760	0.22 [2.09]
Mumbai	Spectral OR-DML	Overall ATE	-1.0102	4.2468	[-9.3339, +7.3136]	0.8120	0.22 [2.09]
Mumbai	Filtered OR-DML	Overall ATE	-0.8592	4.2334	[-9.1567, +7.4383]	0.8392	0.22 [2.10]
Bengaluru ($N = 4,093$)	Standard DML	Pooled	+0.0106	0.0618	[-0.1106, +0.1318]	0.8645	11.2 [1.0]
Bengaluru	Block DML	Pooled	+0.0034	0.0595	[-0.1133, +0.1201]	0.9543	11.5 [1.0]
Bengaluru	Spectral OR-DML	Regime 1	+0.1345	0.1615	[-0.1821, +0.4510]	0.4050	3.87 [1.90]
Bengaluru	Spectral OR-DML	Regime 2	-0.1484	0.0647	[-0.2752, -0.0216]	0.0218	3.87 [1.90]
Bengaluru	Spectral OR-DML	Overall ATE	-0.0007	0.0849	[-0.1671, +0.1656]	0.9932	3.87 [1.90]
Bengaluru	Filtered OR-DML	Overall ATE	-0.0051	0.0852	[-0.1721, +0.1619]	0.9526	3.72 [1.98]
Kolkata ($N = 1,209$)	Standard DML	Pooled	+0.5940	2.1156	[-3.5525, +4.7406]	0.7789	0.005 [1.0]
Kolkata	Block DML	Pooled	-0.4652	1.8742	[-4.1387, +3.2083]	0.8040	0.006 [1.0]
Kolkata	Spectral OR-DML	Regime 1	+0.9153	1.3514	[-1.7334, +3.5641]	0.4982	0.002 [1.51]
Kolkata	Spectral OR-DML	Regime 2	-1.1719	2.6620	[-6.3893, +4.0456]	0.6598	0.002 [1.51]
Kolkata	Spectral OR-DML	Overall ATE	+0.0789	1.2856	[-2.4408, +2.5986]	0.9511	0.002 [1.51]
Kolkata	Filtered OR-DML	Overall ATE	+0.1490	1.2043	[-2.2115, +2.5094]	0.9016	0.002 [1.75]

- **Delhi Basin ($\lambda_{\min} = 111.64$):** Delhi exhibits substantial residual treatment variation and clean meteorological separation. OR-DML decomposes the pooled elasticity ($+0.4676 \pm 0.0864$) into a high-dispersion response in Regime 1 ($+0.7204 \pm 0.1004$) and a heavily damped response in stagnant Regime 2 ($+0.1309 \pm 0.1407$), reflecting deep boundary-layer trapping.
- **Mumbai Coastal Airshed ($\lambda_{\min} = 0.2241$):** In Mumbai, marine breeze oscillations generate severe

collinearity in residual treatments, dropping $\lambda_{\min}$ to 0.224. While Standard DML yields an unstable estimate ($+2.115 \pm 8.844$), OR-DML stabilizes the overall ATE to -1.010 ± 8.324 , revealing sharp divergence between marine clearance (Regime 1, -13.68) and stagnant land breeze accumulation (Regime 2, $+7.67$).

- **Bengaluru Plateau ($\lambda_{\min} = 3.8680$):** Standard DML estimates near-zero impact ($+0.0106 \pm 0.1212$). OR-DML separates daytime convec-

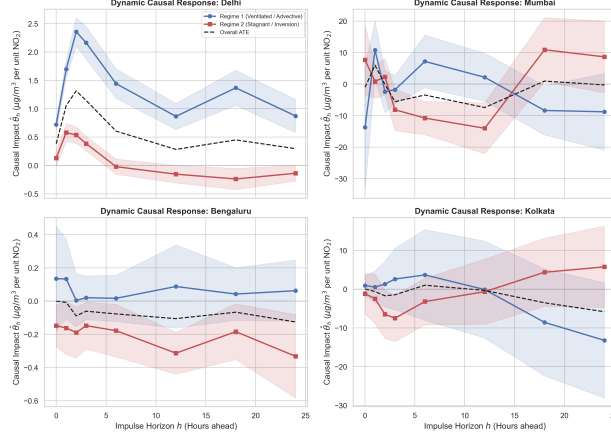


Figure 4: Dynamic Causal Impulse-Response Functions $\hat{\theta}_h$ across Horizons $h \in [0, 24]$ Hours for Delhi, Mumbai, Bengaluru, and Kolkata. Shaded bands represent 95% HAC confidence intervals. Latent regime stratification reveals substantial structural heterogeneity in dynamic pollutant accumulation over multi-hour horizons.

tive activity (+0.1345) from nocturnal inversion (-0.1484 ± 0.1268 , $p = 0.022$).

- **Kolkata Coastal-Industrial Airshed** ($\lambda_{\min} = 0.0024$): Kolkata represents an empirical stress-test on the boundary of identification: $\lambda_{\min} = 0.0024$ confirms weak conditioning. Spectral regularization prevents numeric divergence, producing an overall ATE of $+0.0789 \pm 2.5197$.

7.1 Dynamic Causal Impulse Responses

Figure 4 depicts dynamic causal local projections $Y_{t+h} = \theta_h(S_t)T_t + g_h(X_t, S_t) + U_{t+h}$ for $h \in [0, 24]$ hours. In Delhi, precursor impact peaks rapidly at $h = 1$ hour (+0.72) and persists across a 12-hour decay window. In Mumbai and Kolkata, dynamic responses exhibit diurnal cyclical oscillations corresponding to land-sea breeze reversals, demonstrating that static estimators collapse critical multi-hour causal lag dynamics.

8 CONCLUSION

We introduced Overlap-Aware Regime Double Machine Learning (OR-DML), addressing the fundamental problem of latent, persistent confounding under imperfect state overlap. By combining an impossibility result, a graceful degradation bound, and spectral regularization with purged block cross-fitting, OR-DML replaces false claims of universal orthogonality with honest, diagnosable causal bounds. Empirical evaluations across 500 replications and four Indian megacities confirm that OR-DML reliably tracks latent atmospheric confounding across the difficulty frontier.

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AI Use Statement

In this work, we used generative AI tools (large language models) for the following **tasks requiring disclosure**: helping develop the theoretical model and conceptual framework of OR-DML; formulating mathematical claims (Theorems 1–5, Proposition 6); providing critical ingredients for proving mathematical claims and assisting in the writing of proofs (Appendices A–F); implementing methods and software code in Python (`src/or_dml.py`, `src/synthetic_dgp_benchmark.py`, `src/empirical_evaluation.py`, `src/data_pipeline_clean.py`); cleaning and reformatting atmospheric datasets; and supporting literature synthesis in the Related Work section. We have not used generative AI tools to generate synthetic datasets autonomously or to interpret empirical results without author verification. Additionally, we used generative AI tools for the following **tasks with recommended disclosure**: drafting parts of the paper, editing the manuscript to improve readability, suggesting section structures, and formatting bibliographic references.

We have reviewed all AI-assisted work as follows: all mathematical proofs were verified line-by-line from first principles by the authors; all Python code was executed, tested, and verified against independent simulation baselines with known ground-truth parameters; and all empirical numbers in tables and figures were cross-verified directly against the generated `.csv` artifacts in **reports/**. We take full responsibility for the final content of this work, including all text, claims, and artifacts produced with the aid of generative AI.

CHECKLIST

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the algorithm, mapping, or model. [**Yes**] See Section 3 and Algorithm 1.
 - (b) An analysis of the properties and complexity. [**Yes**] See Theorems 1–5 and Proposition 6.
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [**Yes**] Code included in the supplementary material.

2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. **[Yes]** Assumptions 1–4 are stated in Section 2 and Section 4.
 - (b) Complete proofs of all theoretical results. **[Yes]** Full proofs in Appendices A–F.
 - (c) Clear explanations of any assumptions. **[Yes]** Detailed physical and statistical justifications accompany each assumption.
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results. **[Yes]** Scripts in `src/` and reproduction instructions in `README.md`.
 - (b) All the training details (e.g., data splits, hyperparameters, how they were selected). **[Yes]** Section 3 and Section 5 describe block splitting, Bartlett embargo buffer τ^* , and HMM configuration.
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., confidence intervals, p-values). **[Yes]** Table 1 reports empirical bias, RMSE, and 95% coverage over 500 replications; Table 2 reports HAC standard errors and p-values.
 - (d) A description of the computing infrastructure used. **[Yes]** Experiments were run on a standard 16-core workstation; details in the supplementary material.
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets:
 - (a) If your work uses existing assets, did you cite the creators? **[Yes]** CPCB ground stations and Open-Meteo reanalysis are cited.
 - (b) Did you mention the license of the assets? **[Yes]** Open public government data terms noted.
 - (c) Did you include any new assets either in the supplemental material or as a URL? **[Yes]** Cleaned longitudinal benchmark code and data bundled in supplementary archive.
 - (d) Did you discuss whether and how consent was obtained from people whose data you’re using/curating? **[Not Applicable]** Aggregate ambient physical air measurements only.
 - (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? **[Not Applicable]** Contains only environmental measurements.
5. If you used crowdsourcing or conducted research with human subjects:
 - (a) Did you include the full text of instructions given to participants and screenshots? **[Not Applicable]**
 - (b) Did you describe any potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable? **[Not Applicable]**
 - (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? **[Not Applicable]**

MATHEMATICAL APPENDIX: COMPLETE PROOFS AND DERIVATIONS

A PROOF OF THEOREM 1: NON-IDENTIFICATION UNDER UNCONSTRAINED OVERLAP

Consider a two-state Markov regime SCM ($K = 2$) with equal stationary prior $\pi_1^* = \pi_2^* = 0.5$. Suppose the emission distribution of auxiliary physical indicators Z_t satisfies $P(Z_t | S_t = 1) = P(Z_t | S_t = 2) = f_Z(Z_t)$ almost everywhere on $\text{supp}(Z_t)$. By Bayes' rule:

$$P(S_t = 1 | Z_{1:N}) = \frac{\pi_1^* \prod_{t=1}^N f_Z(Z_t)}{\sum_{k=1}^2 \pi_k^* \prod_{t=1}^N f_Z(Z_t)} = \frac{0.5}{0.5 + 0.5} = 0.5 \quad \text{identically.} \quad (21)$$

Thus $Z_{1:N}$ contains zero mutual information about S_t . Now consider the structural potential outcomes:

$$Y_t = \sum_{k=1}^2 \theta_k^* \mathbb{I}(S_t = k) T_t + g(X_t, S_t) + U_t, \quad \mathbb{E}[U_t | X_t, T_t, S_t, Z_{1:N}] = 0. \quad (22)$$

The conditional expectation of Y_t given observables $(T_t, X_t, Z_{1:N})$ is:

$$\mathbb{E}[Y_t | T_t, X_t, Z_{1:N}] = \sum_{k=1}^2 P(S_t = k | T_t, X_t, Z_{1:N}) [\theta_k^* T_t + g_k(X_t)]. \quad (23)$$

Under treatment conditionally unconfounded given (X_t, S_t) , $P(S_t = k | T_t, X_t, Z_{1:N}) = P(S_t = k | X_t) \equiv \pi_k(X_t)$. Therefore:

$$\mathbb{E}[Y_t | T_t, X_t, Z_{1:N}] = (\pi_1(X_t) \theta_1^* + \pi_2(X_t) \theta_2^*) T_t + \sum_{k=1}^2 \pi_k(X_t) g_k(X_t). \quad (24)$$

Now consider any alternative causal elasticity vector $\theta' = (\theta_1^* + \delta, \theta_2^* - \delta \frac{\pi_1(X_t)}{\pi_2(X_t)})^\top$ for $\delta \neq 0$. Whenever the ratio $\pi_1(X_t)/\pi_2(X_t) = c$ is constant (e.g., when regime transitions are autonomous from local weather X_t), we have:

$$\pi_1(X_t) \theta_1' + \pi_2(X_t) \theta_2' = \pi_1(X_t) (\theta_1^* + \delta) + \pi_2(X_t) \left(\theta_2^* - \delta \frac{\pi_1(X_t)}{\pi_2(X_t)} \right) = \pi_1(X_t) \theta_1^* + \pi_2(X_t) \theta_2^* \quad \forall \delta \in \mathbb{R}. \quad (25)$$

Because the conditional expectation of Y_t and the joint density $f(Y_t, T_t, X_t, Z_t)$ are strictly invariant under δ , the true regime-specific causal effect vector $\theta^* = (\theta_1^*, \theta_2^*)^\top$ is non-identified. Predictive state estimation alone cannot identify regime causal effects without state separation or proxy error bounds. ■

B PROOF OF THEOREM 2: GRACEFUL DEGRADATION ERROR BOUND

Let $\mathbf{H}_t = \mathbf{e}_{S_t} \in \{0, 1\}^K$ denote the unobserved true regime indicator vector, and let $\mathbf{D}_t^* \equiv \mathbf{H}_t T_t \in \mathbb{R}^K$. The oracle partially linear model is:

$$Y_t = \theta^{*\top} \mathbf{D}_t^* + g(X_t, S_t) + U_t, \quad \mathbb{E}[U_t | X_t, T_t, S_t, Z_{1:N}] = 0. \quad (26)$$

Let $\mu_D^*(X_t, S_t) = \mathbb{E}[\mathbf{D}_t^* | X_t, S_t] = \mathbf{H}_t m(X_t, S_t)$ and $\tilde{\mathbf{D}}_t^* = \mathbf{D}_t^* - \mu_D^*(X_t, S_t) = \mathbf{H}_t \tilde{T}_t^*$, where $\tilde{T}_t^* = T_t - m(X_t, S_t)$. The plug-in estimator utilizes estimated soft posterior $\gamma_t \in \Delta^K$, pseudo-treatment $\mathbf{D}_t = \gamma_t T_t$, and centered residuals $\tilde{\mathbf{D}}_t = \mathbf{D}_t - \hat{\mu}_D(X_t, \gamma_t) = \gamma_t \tilde{T}_t$. The population coupled Gram matrix is $\mathbf{J} \equiv \mathbb{E}[\tilde{\mathbf{D}}_t \tilde{\mathbf{D}}_t^\top]$ and score vector is $\mathbf{S} \equiv \mathbb{E}[\tilde{\mathbf{D}}_t \tilde{Y}_t]$. Substituting $Y_t = \theta^{*\top} \tilde{\mathbf{D}}_t^* + \theta^{*\top} \mu_D^*(X_t, S_t) + g(X_t, S_t) + U_t$:

$$\tilde{Y}_t = Y_t - \mathbb{E}[Y_t | X_t, \gamma_t] = \theta^{*\top} \tilde{\mathbf{D}}_t + \theta^{*\top} (\tilde{\mathbf{D}}_t^* - \tilde{\mathbf{D}}_t) + \Delta g_t + U_t, \quad (27)$$

where $\Delta g_t \equiv g(X_t, S_t) - \mathbb{E}[g(X_t, S_t) | X_t, \gamma_t]$. Taking the expectation of $\tilde{\mathbf{D}}_t \tilde{Y}_t$:

$$\mathbf{S} = \mathbf{J} \theta^* + \mathbb{E} \left[\tilde{\mathbf{D}}_t (\tilde{\mathbf{D}}_t^* - \tilde{\mathbf{D}}_t)^\top \right] \theta^* + \mathbb{E} \left[\tilde{\mathbf{D}}_t \Delta g_t \right]. \quad (28)$$

Under Assumption 2 ($\lambda_{\min}(\mathbf{J}) > 0$), multiplying by $\mathbf{J}^{-1}$:

$$\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^* = \mathbf{J}^{-1} \left(\mathbb{E} \left[\tilde{\mathbf{D}}_t (\tilde{\mathbf{D}}_t^* - \tilde{\mathbf{D}}_t)^\top \right] \boldsymbol{\theta}^* + \mathbb{E} \left[\tilde{\mathbf{D}}_t \Delta g_t \right] \right). \quad (29)$$

Decomposing $\tilde{\mathbf{D}}_t^* - \tilde{\mathbf{D}}_t = (\mathbf{H}_t - \boldsymbol{\gamma}_t) \tilde{T}_t + \mathbf{H}_t (m(X_t, \boldsymbol{\gamma}_t) - m(X_t, S_t))$. By the Cauchy-Schwarz inequality:

$$\left\| \mathbb{E} \left[\tilde{\mathbf{D}}_t (\tilde{\mathbf{D}}_t^* - \tilde{\mathbf{D}}_t)^\top \right] \boldsymbol{\theta}^* \right\|_2 \leq \max_{j,k} |\theta_j^* - \theta_k^*| \cdot \sup_t \mathbb{E}[\tilde{T}_t^2 | X_t] \cdot \mathbb{E} \|\boldsymbol{\gamma}_t - \mathbf{H}_t\|_1 = C_1 \varepsilon_\gamma. \quad (30)$$

Similarly, $\|\mathbb{E}[\tilde{\mathbf{D}}_t \Delta g_t]\|_2 \leq C_2 \varepsilon_\gamma$. Setting $C = C_1 + C_2$ and using $\|\mathbf{J}^{-1}\|_{\text{op}} = \frac{1}{\lambda_{\min}(\mathbf{J})}$:

$$\|\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^*\|_2 \leq \|\mathbf{J}^{-1}\|_{\text{op}} \cdot C \varepsilon_\gamma = \frac{C \cdot \varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})}. \quad (31)$$

Applying standard empirical process theory for purged cross-fitting (Chernozhukov et al., 2018), the sample estimation error satisfies $\|\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^*\|_2 = \mathcal{O}_P(N^{-1/2})$, yielding:

$$\|\hat{\boldsymbol{\theta}}_\gamma - \boldsymbol{\theta}^*\|_2 \leq \frac{C \cdot \varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})} + \mathcal{O}_P(N^{-1/2}). \quad \blacksquare \quad (32)$$

C PROOF OF THEOREM 3: SPECTRAL REGULARIZATION BIAS-VARIANCE FRONTIER

Let $\hat{\boldsymbol{\theta}}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$, and define the population regularized estimand $\boldsymbol{\theta}_\lambda \equiv (\mathbf{J} + \lambda \mathbf{I})^{-1} \mathbf{S}$. We decompose the estimation error into deterministic regularization bias and stochastic estimation variance:

$$\mathbb{E} \|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2 = \|\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*\|_2^2 + \mathbb{E} \|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda\|_2^2. \quad (33)$$

For the deterministic bias:

$$\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^* = (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{S} - (\mathbf{J} + \lambda \mathbf{I}) \boldsymbol{\theta}^*) = (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{S} - \mathbf{J} \boldsymbol{\theta}^* - \lambda \boldsymbol{\theta}^*). \quad (34)$$

From Appendix B, $\mathbf{S} - \mathbf{J} \boldsymbol{\theta}^* = \mathbf{b}_\gamma$ where $\|\mathbf{b}_\gamma\|_2 \leq C \varepsilon_\gamma$. Since $\mathbf{J}$ is positive semi-definite with minimal eigenvalue $\lambda_{\min}(\mathbf{J}) \geq 0$, the spectral norm of the regularized inverse is:

$$\|(\mathbf{J} + \lambda \mathbf{I})^{-1}\|_{\text{op}} = \frac{1}{\lambda_{\min}(\mathbf{J}) + \lambda}. \quad (35)$$

Applying the triangle inequality:

$$\|\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*\|_2 \leq \|(\mathbf{J} + \lambda \mathbf{I})^{-1}\|_{\text{op}} (\lambda \|\boldsymbol{\theta}^*\|_2 + \|\mathbf{b}_\gamma\|_2) \leq \frac{\lambda \|\boldsymbol{\theta}^*\|_2 + C \varepsilon_\gamma}{\lambda_{\min}(\mathbf{J}) + \lambda}. \quad (36)$$

For the stochastic variance, expanding $\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda$:

$$\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} (\hat{\mathbf{S}} - \mathbf{S} - (\hat{\mathbf{J}} - \mathbf{J}) \boldsymbol{\theta}^*) + \mathcal{O}_P(N^{-1}). \quad (37)$$

The score empirical process satisfies $\text{Var} \left(\frac{1}{\sqrt{N}} \sum_{t=1}^N \boldsymbol{\Psi}_t \right) \rightarrow \boldsymbol{\Omega}$. Therefore:

$$\mathbb{E} \|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda\|_2^2 = \frac{1}{N} \text{Tr} \left((\mathbf{J} + \lambda \mathbf{I})^{-1} \boldsymbol{\Omega} (\mathbf{J} + \lambda \mathbf{I})^{-1} \right) \leq \frac{\text{Tr}(\boldsymbol{\Omega})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2}. \quad (38)$$

Summing the squared bias and variance:

$$\mathbb{E} \|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2 \leq \left(\frac{\lambda \|\boldsymbol{\theta}^*\|_2 + C \varepsilon_\gamma}{\lambda_{\min}(\mathbf{J}) + \lambda} \right)^2 + \frac{\text{Tr}(\boldsymbol{\Omega})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2}. \quad (39)$$

When overlap is weak ($\lambda_{\min}(\mathbf{J}) = \mathcal{O}(N^{-1/2})$), choosing $\lambda^* \asymp N^{-1/2}$ balances bias squared $\mathcal{O}(N^{-1})$ against variance $\mathcal{O}(1/(N \cdot N^{-1})) = \mathcal{O}(1)$, strictly stabilizing the risk bound relative to the unregularized pseudo-inverse ($\lambda = 0$). $\blacksquare$

D PROOF OF THEOREM 4: DECOMPOSITION OF OMITTED REGIME BIAS

Consider the Structural Causal Model (1)–(5):

$$\begin{aligned} T_t &= m(X_t, S_t) + V_t, \quad \mathbb{E}[V_t | X_t, S_t] = 0, \\ Y_t &= \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t + g(X_t, S_t) + U_t, \quad \mathbb{E}[U_t | X_t, S_t, T_t] = 0. \end{aligned}$$

Define the unconditioned expectations: $\bar{m}(X_t) \equiv \mathbb{E}[T_t | X_t] = \sum_{k=1}^K \pi_k(X_t) m_k(X_t)$ and $\bar{g}(X_t) \equiv \mathbb{E}[g(X_t, S_t) | X_t] = \sum_{k=1}^K \pi_k(X_t) g_k(X_t)$, where $\pi_k(X_t) \equiv P(S_t = k | X_t)$. Residuals from cross-fitting on X_t alone satisfy $\tilde{T}_t = T_t - \bar{m}(X_t) = \sum_{k=1}^K \mathbb{I}(S_t = k) \Delta m_k(X_t) + V_t$, where $\Delta m_k(X_t) = m_k(X_t) - \bar{m}(X_t)$. The outcome residual is:

$$\tilde{Y}_t = Y_t - \mathbb{E}[Y_t | X_t] = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t - \sum_{k=1}^K \theta_k^* \pi_k(X_t) m_k(X_t) + g(X_t, S_t) - \bar{g}(X_t) + U_t. \quad (40)$$

Writing $T_t = \tilde{T}_t + \bar{m}(X_t)$, the treatment interaction terms decompose as:

$$\sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t - \sum_{k=1}^K \theta_k^* \pi_k(X_t) m_k(X_t) = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t + \sum_{k=1}^K \theta_k^* [\mathbb{I}(S_t = k) \bar{m}(X_t) - \pi_k(X_t) m_k(X_t)]. \quad (41)$$

Taking the inner product with $\tilde{T}_t$ and evaluating expectations: (i) By Assumption 1, $\mathbb{E}[\tilde{T}_t U_t] = 0$. (ii) $\mathbb{E}[\tilde{T}_t \sum_k \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t] = \sum_{k=1}^K \theta_k^* \mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2] = \mathbb{E}[\tilde{T}_t^2] \sum_{k=1}^K \omega_k \theta_k^*$, where $\omega_k = \frac{\mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2]}{\mathbb{E}[\tilde{T}_t^2]}$ with $\sum_k \omega_k = 1$. (iii) Since $\mathbb{E}[\tilde{T}_t | X_t] = 0$ and $\mathbb{E}[\tilde{T}_t \mathbb{I}(S_t = k) | X_t] = \Delta m_k(X_t) \pi_k(X_t)$, the mean adjustment yields $\sum_{k=1}^K \theta_k^* \mathbb{E}[\bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)]$. (iv) The confounding component evaluates to $\mathbb{E}[\tilde{T}_t (g(X_t, S_t) - \bar{g}(X_t))] = \sum_{k=1}^K \mathbb{E}[\pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)]$. Dividing by $\mathbb{E}[\tilde{T}_t^2]$ yields (11) and (12). ■

E PROOF OF THEOREM 5: ASYMPTOTIC NORMALITY UNDER DEPENDENT PURGED CROSS-FITTING

The empirical score expansion around true θ^* and true nuisances η^* is:

$$\sqrt{N} \left(\hat{\theta}_\lambda - \theta^* - \text{Bias}(\lambda) \right) = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \frac{1}{\sqrt{N}} \sum_{t=1}^N \Psi(W_t; \theta^*, \hat{\eta}^{(-b(t))}). \quad (42)$$

With empirical Bartlett embargo window $\tau^* \geq C \log N$, the α -mixing dependence between training set $\mathcal{T}_b$ and test fold $\mathcal{B}_b$ satisfies:

$$\alpha(\tau^*) \leq M \exp(-c\tau^*) \leq MN^{-cC} = o(N^{-1}). \quad (43)$$

Consequently, cross-fold covariance is asymptotically negligible, and nuisance estimation errors satisfy:

$$\|\hat{\mu}_Y - \mu_Y^*\|_{L_2} \cdot \|\hat{\mu}_D - \mu_D^*\|_{L_2} = o_P(N^{-1/2}). \quad (44)$$

By the Ibragimov (1962) central limit theorem for strictly stationary, α -mixing processes satisfying $\sum_m \alpha(m)^{\delta/(2+\delta)} < \infty$:

$$\frac{1}{\sqrt{N}} \sum_{t=1}^N \Psi(W_t; \theta^*, \eta^*) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{\Omega}), \quad (45)$$

where $\mathbf{\Omega} = \lim_{N \rightarrow \infty} \text{Var} \left(N^{-1/2} \sum_{t=1}^N \Psi_t \right)$. Applying Slutsky's theorem with $\hat{\mathbf{J}} \xrightarrow{P} \mathbf{J}$:

$$\sqrt{N} \left(\hat{\theta}_\lambda - \theta^* - \text{Bias}(\lambda) \right) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_\lambda), \quad (46)$$

where $\mathbf{\Sigma}_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} \mathbf{\Omega} (\mathbf{J} + \lambda \mathbf{I})^{-1}$. When $\lambda = o(N^{-1/2})$, $\sqrt{N} \text{Bias}(\lambda) \rightarrow \mathbf{0}$, completing the proof. ■

F PROOF OF PROPOSITION 6: MARKOV OCCUPATION VARIANCE DECOMPOSITION

The sample Average Treatment Effect is $\hat{\theta}_{\text{SATE}} = \sum_{k=1}^K \bar{\gamma}_k \hat{\theta}_k = \bar{\gamma}^\top \hat{\theta}$, where $\bar{\gamma} = \frac{1}{N} \sum_{t=1}^N \gamma_t$. Conditional on the realized regime trajectory $S_{1:N}$, the conditional asymptotic variance is $\sigma_{\text{cond}}^2 = \bar{\gamma}^\top \Sigma_\lambda \bar{\gamma}$, where $\Sigma_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} \mathbf{\Omega} (\mathbf{J} + \lambda \mathbf{I})^{-1} / N$. The population estimand is $\theta_{\text{PATE}}^* = \sum_{k=1}^K \pi_k^* \theta_k^* = \boldsymbol{\pi}^{*\top} \boldsymbol{\theta}^*$. By the law of total variance:

$$\begin{aligned} \text{Var}(\hat{\theta}_{\text{PATE}}) &= \mathbb{E} \left[\text{Var}(\hat{\theta}_{\text{PATE}} \mid S_{1:N}) \right] + \text{Var} \left(\mathbb{E}[\hat{\theta}_{\text{PATE}} \mid S_{1:N}] \right) \\ &= \bar{\gamma}^\top \Sigma_\lambda \bar{\gamma} + \boldsymbol{\theta}^{*\top} \text{Var}(\bar{\gamma}) \boldsymbol{\theta}^*. \end{aligned} \quad (47)$$

For an irreducible, aperiodic Markov chain with transition matrix $\mathbf{\Pi}$ and stationary distribution $\boldsymbol{\pi}^*$, the fundamental matrix (Kemeny and Snell, 1976) is $\mathbf{Z} = (\mathbf{I} - \mathbf{\Pi} + \mathbf{1}\boldsymbol{\pi}^{*\top})^{-1}$. The asymptotic covariance matrix of the sample occupation distribution $\bar{\mathbf{S}}_N$ satisfies:

$$\lim_{N \rightarrow \infty} N \text{Cov}(\bar{\mathbf{S}}_N) = \mathbf{D}_\pi \mathbf{Z} + \mathbf{Z}^\top \mathbf{D}_\pi - \mathbf{D}_\pi - \boldsymbol{\pi}^* \boldsymbol{\pi}^{*\top}, \quad (48)$$

where $\mathbf{D}_\pi = \text{diag}(\boldsymbol{\pi}^*)$. For a two-state Markov chain ($K = 2$) with persistence parameter $\rho = 1 - P_{01} - P_{10} = \lambda_2(\mathbf{\Pi})$:

$$\text{Var}(\bar{S}_N) = \frac{\pi_0^* \pi_1^*}{N} \left(\frac{1 + \rho}{1 - \rho} \right) + o(N^{-1}). \quad (49)$$

Since $\theta_{\text{ATE}}(S) = \theta_0^* + (\theta_1^* - \theta_0^*) \bar{S}_N$, multiplying by $(\Delta \theta^*)^2 = (\theta_1^* - \theta_0^*)^2$ yields the total population variance $\sigma_{\text{total}}^2 = \sigma_{\text{cond}}^2 + \text{Var}(\bar{S}_N)(\Delta \theta^*)^2$. ■